

spikformer+Spik4lite Demo

Introduction

This is a demo implementation of "**Spik4lite: Refactoring Neuromorphic Sparsity for Efficient Spiking Neural Networks on Commodity Edge Devices**". The demo show the basic usage of spikformer+Spik4lite.

Directory

The `spikformer` is the root and its structure is described as below.

```
spikformer/
├── README.md
├── cifar10
│   ├── aa_snn.py
│   ├── cifar10.yml
│   ├── loader.py
│   ├── model.py
│   ├── test.py
│   ├── train.py
│   └── transforms_factory.py
├── cifar10dvs
│   ├── autoaugment.py
│   ├── model.py
│   ├── test.py
│   ├── train.log
│   ├── train.py
│   └── utils.py
├── cifar100
│   ├── aa_snn.py
│   ├── cifar100.yml
│   ├── loader.py
│   ├── model.py
│   ├── test.py
│   ├── train.log
│   ├── train.py
│   └── transforms_factory.py
└── dvs128gesture
    ├── autoaugment.py
    ├── model.py
    ├── test.py
    ├── train.py
    └── utils.py
```

- * `README.md` provides the general overview and introduction of the project, as well as instructions on how to run the code.

Running the Demo

The whole pipeline contains three major steps: (1) activate the Conda Environment, (2) Run the train code.

(1) activate the Conda Environment

```
cd spikformer
conda activate Spik4lite
```

(2) Run the code

Runing on CIFAR10

Setting hyper-parameters in cifar10.yml

```
cd cifar10  
python train.py
```

Runing on CIFAR100

Setting hyper-parameters in cifar100.yml

```
cd cifar100  
python train.py
```

Runing on DVS128 Gesture

```
cd dvs128gesture  
python train.py
```

Runing on CIFAR10-DVS

```
cd cifar10dvs  
python train.py
```